

Dynamics of quantum particles by path-integral centroid simulations: The symmetric Eckart barrier

Rafael Ramírez

Citation: *The Journal of Chemical Physics* **107**, 3550 (1997); doi: 10.1063/1.474694

View online: <http://dx.doi.org/10.1063/1.474694>

View Table of Contents: <http://scitation.aip.org/content/aip/journal/jcp/107/9?ver=pdfcov>

Published by the [AIP Publishing](#)

Articles you may be interested in

[Variational R-matrix method for quantum tunneling problems](#)

Am. J. Phys. **71**, 273 (2003); 10.1119/1.1519231

[The Schrödinger formulation of the Feynman path centroid density](#)

J. Chem. Phys. **111**, 3339 (1999); 10.1063/1.479666

[General solution scheme for second-order differential equations: application to quantum transport](#)

Comput. Phys. **12**, 248 (1998); 10.1063/1.168679

[Evaluation of coherent-state path integrals in statistical mechanics by matrix multiplication](#)

J. Chem. Phys. **108**, 1562 (1998); 10.1063/1.475527

[Optimal energy estimation in path-integral Monte Carlo simulations](#)

J. Chem. Phys. **107**, 5821 (1997); 10.1063/1.474309



Dynamics of quantum particles by path-integral centroid simulations: The symmetric Eckart barrier

Rafael Ramírez

Instituto de Ciencia de Materiales, Consejo Superior de Investigaciones Científicas (C.S.I.C.), Cantoblanco, 28049 Madrid, Spain

(Received 4 March 1997; accepted 27 May 1997)

The path-integral centroid approach has been applied to study the dynamical properties of a flux of protons impinging on a symmetric Eckart barrier. The mean transmission coefficient, transmitted flux, and kinetic energy of transmitted particles have been calculated by path-integral centroid simulations as a function of temperature, and compared to exact results obtained from the solution of the Schrödinger equation. The studied temperatures cover the crossover from a classical regime, where the barrier crossing is thermally activated, to a quantum regime, where the barrier crossing is dominated by tunneling of low energy particles. We show, in agreement with previous studies, that the centroid density is a central quantity to derive dynamical properties. Moreover, we find that the equilibrium internal energy obtained for the centroid fixed at the barrier top, reproduces closely the difference between the mean kinetic energy of transmitted and incident particles, and it can be used to define a velocity (pre-exponential) factor that improves previous approximations to the transmitted flux, in the whole temperature range above and below the classical-quantum crossover.

© 1997 American Institute of Physics. [S0021-9606(97)00233-X]

I. INTRODUCTION

In the application of the path-integral (PI) formulation to the statistical mechanics of quantum systems, Feynman introduced the notion of the centroid of quantum paths to represent the partition function of the quantum system by a classical formula.¹ The use of the centroid as the relevant coordinate in the study of the dynamics of quantum systems was first suggested by Gillan,^{2,3} showing that the barrier-crossing rate for a particle in a symmetric well coupled to a sluggish bath can be approximated by a generalization of classical transition-state theory (TST), whose key element was the ratio of probabilities for finding the centroid at the barrier top and at the stable site. The centroid of the quantum paths appears then as a good candidate to represent static and dynamical properties of quantum particles in thermodynamic equilibrium by a classical coordinate.

A large effort has been devoted to the rigorous formulation of Gillan's approach as the so-called quantum transition-state theory.^{4,5} Voth demonstrated the way in which a centroid dividing surface appears in this theory.^{5,6} In a large series of papers, Cao and Voth⁷ introduced general rules for the calculation of equilibrium and dynamical properties of quantum systems as operator averages and imaginary time correlation functions, using the centroid density as the central statistical distribution. Perhaps their most interesting result is the approximate method called "centroid molecular dynamics" employed for computing quantum dynamical time correlation functions. Makarov and Topaler demonstrated that the centroid of the quantum paths provides a proper choice of the unstable mode in TST only if the potential is weakly asymmetric or if the temperature is high enough.⁸ Moreover, for a metastable potential not bounded from below the centroid density diverges if the potential decreases not too slowly. A justification for the introduction of

the centroid variable has been presented by Cao and Voth in their unified framework for quantum activated dynamics.⁹ Nevertheless, the limits of the validity of path-integral centroid approaches are not clearly established for a general potential. In this context, computer simulations may help to understand the capability and limitations of the method.

The main advantage of the path-integral centroid approach is that it allows us to study dynamical effects as a function of temperature in many-body systems, where the relevant degrees of freedom must be treated by quantum mechanics in regions at which a harmonic approximation fails. Most applications have been centered in studies of activated quantum processes in condensed matter systems, as the diffusion of hydrogen in metals, both in the bulk¹⁰ and at the surface,^{11,12} and in crystalline pure¹³ and doped¹⁴ semiconductors. These investigations included the treatment of several quantum degrees of freedom and many more classical ones, and a central issue was to describe the crossover from a classical thermally activated diffusion to a low temperature regime dominated by quantum tunneling. Typical crossover temperatures (T_c) in these systems are found in a range between 250 K and 50 K. Such a crossover has been found even in reactions involving biological molecules, as in the rate of CO migration to a separated β chain of hemoglobin, with a T_c around 25 K.¹⁵ Other results based on path-integral centroid simulations dealing with proton transfer in polar solvents¹⁶ and heterogeneous electron transfer¹⁷ have recently been reviewed by Voth.⁵

As the path-integral centroid technique is becoming a standard approach to studying the quantum dynamics of highly complex systems, it is important to test this technique in simple models, where a comparison to exact results is possible. However, the number of such studies is small so far. The one-dimensional reactive flux of protons in a sym-

metric Eckart barrier has been studied down to temperatures below the crossover.^{4,9} In his study of the quartic double well coupled to a sluggish bath, Gillan showed that for low temperatures his centroid expression reproduces the instanton solution for the rate constant.² A general relation of the centroid and instanton theories has been presented in Refs. 4, 5, 9. A reaction model corresponding to a single solute coordinate coupled to a harmonic bath mode, described by either one or ten oscillators, was investigated by accurate quantum-mechanical calculations, and compared to the path-integral centroid method at two temperatures, to check the effect of friction in the reactive flux.¹⁸ In the limit of low friction the centroid approach deviates from the accurate quantum result by a factor of two at low temperatures.

In the present investigation we aim at improving the understanding of the path-integral centroid method by performing a numerical investigation of the dynamical properties of a flux of protons impinging on a symmetric Eckart barrier at temperatures above and below the classical–quantum crossover. The advantage of this simple model is that exact analytical solutions are readily available by a solution of the Schrödinger equation. This was the first realistic type of barrier for which the tunneling problem was solved,¹⁹ and the results have been extensively used to test theories dealing with chemical reactivity.²⁰ In comparison to previous PI centroid simulations on this model,^{4,9} the range of studied temperatures is considerably extended in the present work, where the lowest temperatures are ten times smaller than T_c . Besides the normalized transmitted flux, we focus on other properties that have not been previously studied by the PI centroid approach, as the mean transmission coefficient of the barrier and the mean kinetic energy of transmitted particles. Moreover, it is shown that not only the centroid density is a relevant quantity in the study of the dynamical properties of this system, but also the internal energy corresponding to the centroid coordinate fixed at the barrier top has a plain physical meaning. This energy turns out to be important to define an approximate velocity correction to the transmitted flux that works better than other existing approximations. The paper is organized as follows. In Sec. II, the model to be studied is presented, and those quantities that will be compared to the simulation results are defined. Some technical details about the path-integral centroid simulations are presented in Sec. III. The simulation data are compared to exact values in Sec. IV, that includes also a discussion of the results. The paper is closed with a summary.

II. PHYSICAL MODEL

The physical model to be investigated corresponds to a steady state in one dimension, in which an incident flux of protons has a distribution of initial velocities given by the equilibrium Maxwell–Boltzmann formula:

$$\rho(v) = \left(\frac{2\beta m}{\pi} \right)^{1/2} \exp\left(-\frac{\beta m v^2}{2} \right), \quad (1)$$

where β is the inverse temperature $(k_B T)^{-1}$, k_B being the Boltzmann constant, and m is the proton mass ($m = 1836$

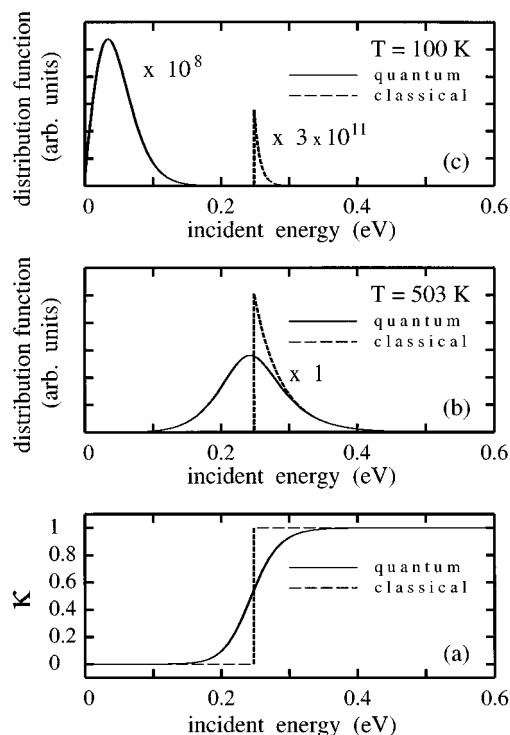


FIG. 1. (a) Transmission coefficient, κ , for a proton in the studied symmetric Eckart barrier as a function of the initial kinetic energy of the particle. Both the quantum result and the classical limit are shown. (b) Quantum and classical distribution functions of transmitted protons as a function of the incident energy at $T = 503$ K. (c) Distribution functions at 100 K. Note the magnification factor used to plot the curves at 100 K in comparison to those shown at 503 K.

a.u.). $\rho(v)$ is normalized to unity when integrated over the positive velocities. The protons move towards a symmetric Eckart barrier $V(x)$, whose barrier top is located at $x = 0$:

$$V(x) = V_b \operatorname{sech}^2(x/a). \quad (2)$$

The parameters defining the potential are the barrier height $V_b = 0.2485$ eV and the distance $a = 0.3491$ Å. These values have been chosen to correspond to the same physical situation as that studied in Refs. 4, 9. The barrier height and width are typical values encountered, for example, in the diffusion of H in crystalline silicon.¹³ The assumption of a steady state implies that the flux of particles $j(v)$ with initial velocity v is proportional to their velocity, $j(v) = Cv$, where the constant C , with dimension of number of particles per unit length, is independent of v .

The transmission coefficient for an incident energy $e = (1/2)mv^2$ can be readily obtained by solving the Schrödinger equation for the symmetric Eckart potential.¹⁹

$$\kappa(e) = \frac{\cosh[2\alpha(e/V_b)^{1/2}] - 1}{\cosh[2\alpha(e/V_b)^{1/2}] + \cosh[(4\alpha^2 - \pi^2)^{1/2}]}, \quad (3)$$

with $\alpha = 2\pi V_b / \hbar v_b$ and $v_b = (1/2\pi)(|V''(0)|/m)^{1/2}$. $V''(0)$ is the second derivative of the Eckart potential at $x = 0$. For the employed potential and particle mass, one has $\alpha = 12$. In Fig. 1(a), we show the transmission probability $\kappa(e)$ as a function of the incident energy of the proton, together with

its classical limit. It is interesting to compare the distribution function of transmitted protons as a function of the incident energy in both quantum and classical cases. At temperatures above the classical–quantum crossover ($T_c = 140$ K, see below), the energy corresponding to the maximum in the distribution function is similar in both cases, as shown in Fig. 1(b) for $T = 503$ K. However, below the crossover temperature, the typical energy of the transmitted particles in the quantum case is much lower than in the classical one, due to the tunneling of the low energy incident protons across the barrier. In Fig. 1(c), we display the distributions obtained at 100 K. Note the different scale used to represent the distribution functions at $T = 100$ K.

Any dynamical quantity, $f(v)$, can be expressed as an average over the velocity distribution of the incident protons:

$$\langle f \rangle = \int_0^\infty dv \rho(v) f(v). \quad (4)$$

The mean initial kinetic energy E_i is the ratio between the incident energy flux to the incident particle flux:

$$E_i = \frac{m}{2} \frac{\langle v^3 \rangle}{\langle v \rangle} = k_B T. \quad (5)$$

The mean transmission coefficient is $K = \langle \kappa \rangle$, and J will be the ratio of the transmitted particle flux to the incident flux, which will be called the normalized transmitted flux:

$$J = \frac{\langle v \kappa \rangle}{\langle v \rangle}. \quad (6)$$

The mean kinetic energy of the transmitted particles is given by

$$E_t = \frac{m}{2} \frac{\langle v^3 \kappa \rangle}{\langle v \kappa \rangle}. \quad (7)$$

As we want to compare properties of a steady state, with results obtained in thermal equilibrium by PI centroid simulations, we will artificially treat the normalized transmitted flux, J , as a partition function, that defines a free energy, F , and an internal energy, E , by the equilibrium formulas:

$$J = \exp(-\beta F), \quad (8)$$

$$E = -\frac{\partial \ln J}{\partial \beta} = E_t - E_i. \quad (9)$$

The last equality follows from the definition given in Eq. (6) after straightforward calculations. In the classical limit, the following expressions for the quantities K , J , and E are obtained:

$$K_{cla} = \text{erfc}[(\beta V_b)^{1/2}], \quad (10)$$

$$J_{cla} = \exp(-\beta V_b), \quad (11)$$

$$E_{cla} = V_b. \quad (12)$$

Note that the classical normalized transmitted flux has the form of a Boltzmann factor. The function erfc is the complementary of the error function, defined as

$$\text{erfc}(x) = 1 - \text{erf}(x) = \frac{2}{\pi^{1/2}} \int_x^\infty \exp(-t^2) dt. \quad (13)$$

In the quantum case it is not possible to obtain closed analytical expressions for these magnitudes,²¹ and they must be evaluated by solving numerically the integrals defined in Eq. (4). I have used standard software for performing this numerical integration.²² The quantum correction factor Γ is the ratio of the quantum to the classical transmitted flux, which is written as

$$\Gamma = J/J_{cla} = \exp(\beta V_b) J. \quad (14)$$

III. PATH-INTEGRAL CENTROID SIMULATIONS

The path-integral representation of the equilibrium partition function, Z , of a quantum particle in an external potential $V(x)$ is formulated as

$$Z = \int_{x(0)=x(\beta\hbar)} Dx(\tau) \exp\left(-\frac{S[x(\tau)]}{\hbar}\right). \quad (15)$$

Only those paths which satisfy the boundary condition $x(0) = x(\beta\hbar)$, i.e., which are periodic functions with period $\beta\hbar$, contribute to Z at a given inverse temperature β . $S[x(\tau)]$ is the action functional of a thermal path $x(\tau)$:

$$S[x(\tau)] = \int_0^{\beta\hbar} d\tau \left(\frac{m}{2} \dot{x}(\tau)^2 + V[x(\tau)] \right). \quad (16)$$

In the discretized path-integral representation, the paths $x(\tau)$ are divided into N steps, and the partition function for a quantum particle is given by the expression¹

$$Z = \lim_{N \rightarrow \infty} \left(\frac{mN}{2\pi\beta\hbar^2} \right)^{N/2} \int dx_1 \cdots \int dx_N \times \exp\left(-\frac{1}{\hbar} \sum_{p=1}^N \left[\frac{mN}{2\beta\hbar} (x_{p+1} - x_p)^2 + \frac{\beta\hbar}{N} V(x_p) \right] \right), \quad (17)$$

where the cyclic condition for the discretized paths is formulated by $x_{N+1} = x_1$. The unnormalized centroid density is given by the constrained path integral:

$$\rho(x_c) = \int_{x(0)=x(\beta\hbar)} Dx(\tau) \delta(x_c - \bar{x}) \exp\left(-\frac{S[x(\tau)]}{\hbar}\right), \quad (18)$$

where $\bar{x}$ is the centroid of a cyclic quantum path $x(\tau)$ defined as

$$\bar{x} = \frac{1}{\beta\hbar} \int_0^{\beta\hbar} d\tau x(\tau). \quad (19)$$

The relation between the partition function, Z , and the centroid density, $\rho(x_c)$, is given by

$$Z = \int dx_c \rho(x_c). \quad (20)$$

The centroid density is used to define the so-called effective classical potential,²³ F_c , that as it depends on temperature, it plays a role analogue to a free energy:

$$Z_c = \exp(-\beta F_c) = \left(\frac{2\pi\beta\hbar^2}{m} \right)^{1/2} \rho(x_c). \quad (21)$$

Z_c is called the local partition function.²⁴ The equilibrium internal energy, E_c , for a fixed centroid position is given by $-\partial(\ln Z_c)/\partial\beta$. Using a discretized representation of the path integral, one obtains

$$E_c(x_c) = \frac{1}{2} k_B T (N-1) - \left\langle \sum_{p=1}^N \left[\frac{mN}{2\beta^2\hbar^2} (x_{p+1} - x_p)^2 - \frac{1}{N} V(x_p) \right] \right\rangle_{x_c}. \quad (22)$$

The symbol $\langle \dots \rangle_{x_c}$ indicates a path-integral average over thermal paths with the centroid fixed at x_c . The usual prescription to evaluate F_c is by performing a line integral of the mean force, $f(\bar{x})$, acting upon the thermal paths:^{2,3}

$$F_c(x_c) - F_c(-\infty) = \int_{-\infty}^{x_c} d\bar{x} f(\bar{x}), \quad (23)$$

$$f(\bar{x}) = - \left\langle \frac{dV}{d\bar{x}} \right\rangle_{\bar{x}}. \quad (24)$$

The integral in Eq. (23) represents the work done on the quantum particle when it is moved in the configuration space defined by the centroid coordinate $\bar{x}$, from $-\infty$ to the position x_c . The relevant quantities derived from the PI centroid simulations to be compared to the steady state problem presented in the previous section are the following energy differences:

$$F^* = F_c(0) - F_c(-\infty), \quad (25)$$

$$E^* = E_c(0) - E_c(-\infty). \quad (26)$$

For the employed Eckart barrier, the conditions $F_c(-\infty) = E_c(-\infty) = 0$ are fulfilled at all temperatures, therefore F^* and E^* correspond to the effective potential and the internal energy when the centroid is fixed at the barrier top.

The discretization employed in the present PI centroid simulations was temperature dependent with $N \approx 10000/T$ K⁻¹. The mean force, at given T and x_c , was obtained by Monte Carlo (MC) simulation using 8×10^5 Monte Carlo steps (MCS), which were preceded by 4×10^5 MCS for equilibration. A MCS involves an attempt to move each of the N discrete coordinates used to represent a thermal path. We have calculated $F_c(x_c)$ at 17 different temperatures in the range between 10000 K and 27 K. The integration in Eq. (23) was performed numerically evaluating the mean force at about 100 different positions of the centroid coordinate. An interval of 0.01 Å was chosen for those regions showing the largest variation in the mean force. The estimated error in F_c

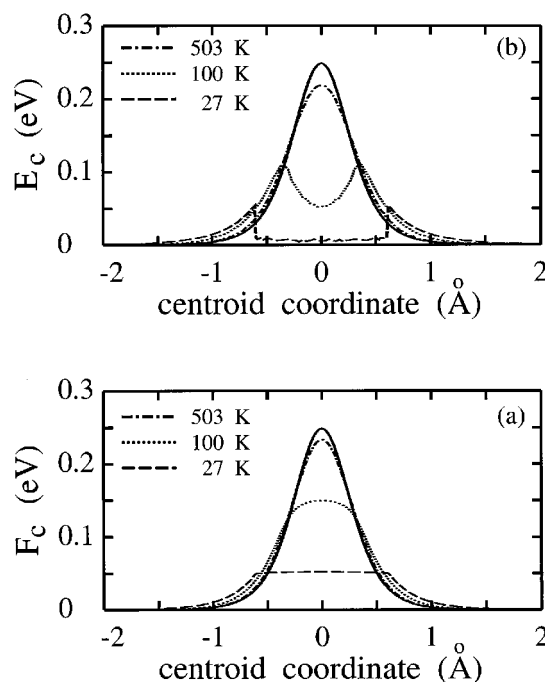


FIG. 2. (a) Effective potential, F_c , as a function of the centroid coordinate at three different temperatures. (b) Internal energy E_c as a function of the centroid position. The continuous line in both panels represents the symmetric Eckart potential.

under these computational conditions is less than 1 meV. A second series of simulations was performed for 80 temperatures in the interval between 10000 K and 10 K with the centroid coordinate fixed at the barrier top, i.e., $x_c = 0$. The thermal averages were calculated over 5.6×10^6 MCS after 1.2×10^6 MCS for equilibration.

In Fig. 2 we display the values obtained for the effective potential F_c and the internal energy E_c at three different temperatures as a function of the centroid position x_c . For reference, the symmetric Eckart potential $V(x)$ is shown by a continuous line. The height of the barrier for the effective potential, F_c , diminishes with temperature and the barrier top is found at $x_c = 0$, independently of the temperature. At low temperatures the effective potential is almost flat in the barrier region of the external potential.^{2,8} The behavior of the internal energy is more unexpected. At $x_c = 0$ and high temperatures, E_c displays an absolute maximum, while at low temperatures there appears an energy minimum. At 27 K, the function $E_c(x_c)$ decreases abruptly at x_c around ± 0.6 Å.

IV. RESULTS AND DISCUSSION

In Fig. 3, the energy E , which gives the difference between the mean kinetic energy of transmitted and incident protons as defined in Eq. (9), is compared to the energy E^* . The temperature dependence of both energies agrees closely. The high temperature limit of E , i.e., V_b [see Eq. (12)], is correctly reproduced by E^* , and it seems that the low temperature limit is also correctly given. At intermediate temperatures, the E^* values are systematically greater than those corresponding to E . The classical–quantum crossover is ac-

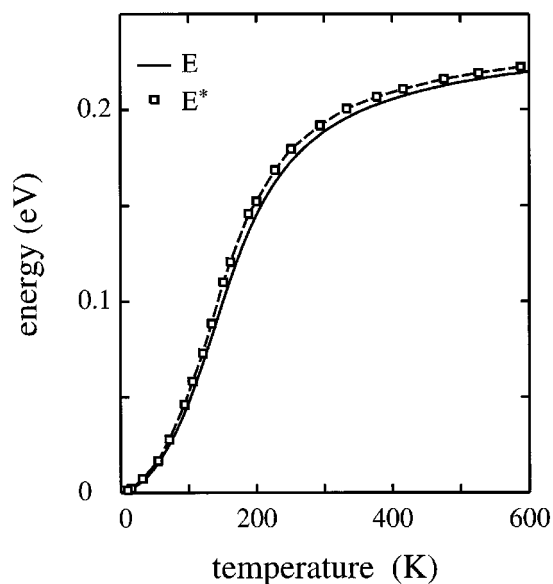


FIG. 3. The energy E , which is the difference between the mean kinetic energy of transmitted and incident protons, is compared to the internal energy E^* , derived from path-integral simulations with the centroid coordinate fixed at the barrier top, at several temperatures. The high temperature limit of both E and E^* is the barrier height V_b .

accompanied by a fast decrease in the energy E as the temperature goes down, since the tunneling of low energy particles becomes dominant. Therefore, we define the crossover temperature, T_c , by the maximum in the temperature derivative of E . For the studied Eckart barrier, this temperature corresponds to $T_c = 140$ K. A parabolic approximation at the barrier top would predict a much higher crossover temperature of about 240 K.

The energy distributions corresponding to transmitted energy flux and to the equilibrium thermal paths are presented in Fig. 4 for two temperatures above and below the classical–quantum crossover. Both distributions are associated to different physical situations. On one side, the distribution of transmitted energy flux corresponds to a steady state generated by a Maxwell–Boltzmann distribution of incident protons. The energy distribution of thermal paths corresponds to a particle in thermal equilibrium with the constraint that the path centroid is fixed at the barrier top. However, the mean energy associated to both distributions, E_t and E^* , are connected by a simple empirical relation: $E = E_t - k_B T \approx E^*$. The distribution of thermal paths may display finite values at negative energies. They correspond to paths that are highly delocalized, so that the leading contribution to their internal energy is given by the harmonic energy, i.e., second term in the r.h.s of Eq. (22), which, due to the sign in front of it, takes finite values that are always lower than zero.

The effective potential, F^* , is compared to the energy F derived from the normalized transmitted flux, J , in Fig. 5. There appears a systematic deviation of F^* towards higher values at intermediate temperatures. The high temperature limit of F and F^* corresponds to the value V_b . The overall agreement between both quantities is remarkable, even for

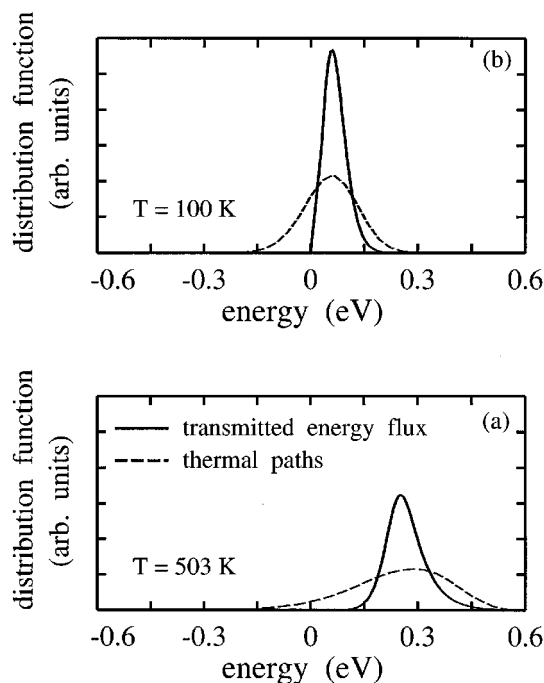


FIG. 4. The continuous line shows the distribution of transmitted energy flux as a function of the incident energy of the protons. The broken line displays the distribution function of equilibrium thermal paths of a proton as a function of the internal energy, when the centroid coordinate is fixed at the barrier top. The results are shown for temperatures of 503 K (a) and 100 K (b).

the lowest studied temperature $T = 10$ K. It is interesting to realize that since the PI centroid approach deals with equilibrium states associated to the local partition function, Z_c [see Eq. (21)], there is an exact thermodynamical relation

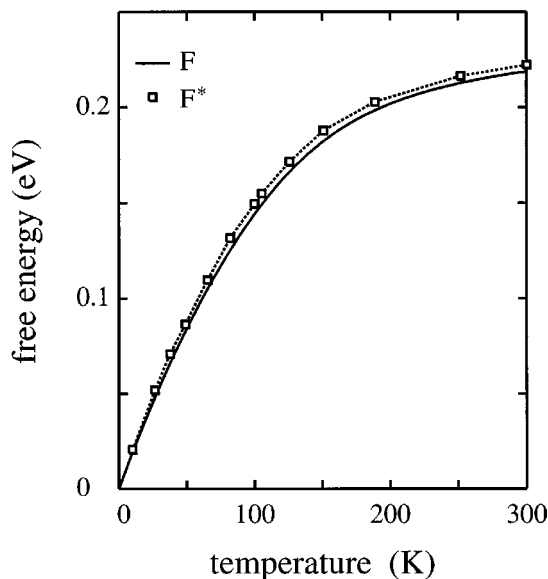


FIG. 5. The temperature dependence of the effective-potential barrier, F^* , obtained by PI centroid simulations, is compared to the energy F , derived from the exactly calculated normalized transmitted flux. The high temperature limit of F and F^* is the barrier height V_b .

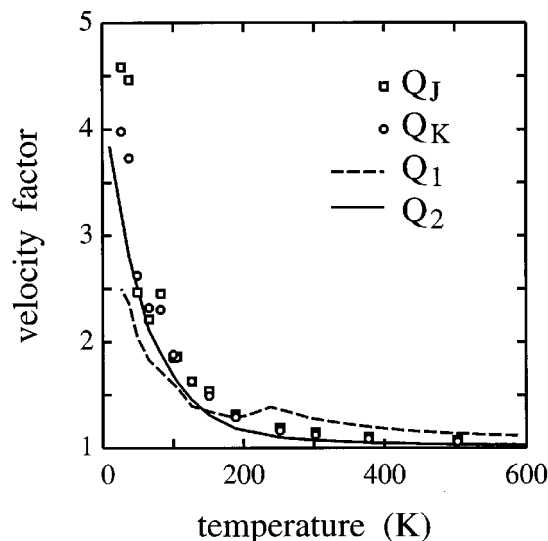


FIG. 6. Temperature dependence of the velocity factors Q_J and Q_K . Q_1 and Q_2 are approximations to the velocity factor, Q_J , discussed in the text.

between the free energy F^* and the internal energy E^* given by

$$E^* = \frac{\partial(\beta F^*)}{\partial \beta}. \quad (27)$$

This means that the free energy F^* can be derived by thermodynamical integration of the previous equation, if the internal energy E^* is calculated for an appropriate set of temperatures. Note that, for the Eckart barrier, the determination of E^* at a given temperature implies just one simulation with the centroid fixed at the barrier top. For a general potential, without the asymptotic behavior of the Eckart barrier at infinity, the calculation of E^* would imply two PI simulations with the centroid fixed at the barrier and at the reactant regions. The calculation of F^* at the same set of temperatures by line integrals of mean force, would need a much larger computational effort, as the numerical evaluation of the line integral at a given temperature requires more than the two PI simulations needed to obtain E^* . Then, the numerical integration of Eq. (27) provides an efficient procedure to compute effective-potential barriers, whose capability will be presented in a separate work.²⁵ The value of F^* shown in Fig. 5 for the lowest temperature $T=10$ K was evaluated in this way. A different procedure envisaged in Ref. 4 for the calculation of F^* , which is based on the determination of the excess free energy with respect to a parabolic barrier, is also efficient in the sense that, for the Eckart barrier, it requires only PI simulations with the centroid constrained at the barrier top.

We turn now to compare the exact mean transmission coefficient, K , and the normalized transmitted flux, J , with those values derived from the PI centroid simulations. The standard approximation to J , known as quantum transition-state theory, is just to factorize this magnitude into an equilibrium part, given by $J^* = \exp(-\beta F^*)$, which carries the most important part of the temperature dependence of the

TABLE I. Quantum correction factors to the symmetric Eckart barrier. Γ is the exact value, $\Gamma^* = \exp(\beta V_b) \exp(-\beta F^*)$, $\Gamma_1^* = Q_1 \Gamma^*$, and $\Gamma_2^* = Q_2 \Gamma^*$.

T (K)	Γ	Γ_2^*	Γ_1^*	Γ^*
10000	1.00	1.00	1.00	1.00
833.33	1.18	1.16	1.23	1.14
503.27	1.52	1.47	1.62	1.42
377.46	2.07	1.97	2.25	1.88
301.96	3.10	2.90	3.45	2.70
251.64	5.20	4.80	5.97	4.37
188.73	21.8	19.6	21.3	16.6
150.98	161.9	137.7	141.3	105.3
125.82	1973.4	1769.0	1692.5	1212.6
105.26	56983	49903	47977	30582
81.97	3.77×10^7	2.94×10^7	2.65×10^7	1.54×10^7
48.78	1.34×10^{17}	1.37×10^{17}	1.12×10^{17}	5.44×10^{16}
26.60	7.84×10^{37}	5.58×10^{37}	4.28×10^{37}	1.71×10^{37}

transmitted flux, and a prefactor, the so-called velocity factor Q_J , which depends weakly on the temperature:

$$J = Q_J \exp(-\beta F^*) = Q_J J^*. \quad (28)$$

This formula agrees, apart from the velocity factor Q_J , with the classical result, J_{cla} , after substitution of the energy barrier V_b in Eq. (11) by the effective-potential barrier F^* . We can apply the same classical correspondence to the mean transmission coefficient K , and substitute V_b by F^* in Eq. (10), obtaining the following relation:

$$K = Q_K^2 \text{erfc}(\beta F^*)^{1/2} = Q_K^2 K^*, \quad (29)$$

which serves to define Q_K as a correction factor. The factors Q_J and Q_K obtained from Eqs. (28) and (29) are shown in Fig. 6 as a function of temperature. Both factors are, within the statistical uncertainty, identical. At high temperature, they converge towards one, while at temperatures below the crossover the velocity factor increases and takes a value of ~ 4 at temperatures about 25 K.

The estimation for the velocity factor, Q_J , presented by Gillan,² and later modified by Cao and Voth,^{9,4} is given by

$$Q_1 = 2\pi\nu \left(\frac{m}{|F_c''(0)|} \right)^{1/2}, \quad (30)$$

where $F_c''(0)$ is the second derivative of the effective potential with respect to the centroid coordinate evaluated at the barrier top $x_c=0$. The frequency ν is given by

$$\nu = \frac{\omega_b}{2\pi}, \quad \text{if } \hbar\beta\omega_b < 2\pi, \quad (31)$$

$$\nu = \frac{1}{\hbar\beta}, \quad \text{if } \hbar\beta\omega_b \geq 2\pi, \quad (32)$$

where $\omega_b = 2\pi\nu_b$, ν_b being defined immediately after Eq. (3). I have derived a new approximation to the velocity factor, based on an empirical relation, for which I have not found any sound justification:

$$\left(\frac{2E}{m} \right)^{1/2} J \approx \left(\frac{2F^*}{m} \right)^{1/2} J^*. \quad (33)$$

TABLE II. Normalized transmitted flux of the symmetric Eckart barrier. J is the exact value, $J^* = \exp(-\beta F^*)$, and $J_2^* = Q_2 J^*$.

T (K)	J	J_2^*	J^*
10000	0.75	0.75	0.75
503.27	4.95×10^{-3}	4.78×10^{-3}	4.62×10^{-3}
301.96	2.21×10^{-4}	2.07×10^{-4}	1.93×10^{-4}
188.73	5.04×10^{-6}	4.53×10^{-6}	3.84×10^{-6}
125.82	2.20×10^{-7}	1.97×10^{-7}	1.35×10^{-7}
81.97	1.99×10^{-8}	1.55×10^{-8}	0.81×10^{-8}
48.78	2.85×10^{-9}	2.91×10^{-9}	1.16×10^{-9}
26.60	6.42×10^{-10}	4.57×10^{-10}	1.40×10^{-10}

Replacing the energy E by E^* (see Fig. 3), the following approximation to the velocity factor is obtained:

$$Q_2 = \left(\frac{F^*}{E^*} \right)^{1/2}.$$

(34)

The two approximations, Q_1 and Q_2 , to the velocity factor, Q_J , are shown in Fig. 6. Q_1 has a discontinuity in its first derivative with respect to temperature at 240 K, that corresponds to the crossover temperature of a parabolic barrier with the same curvature as that of the Eckart potential at the barrier top. At temperatures above 240 K, Q_1 overestimates the value of Q_J , and the opposite behavior is encountered at temperatures below 240 K. The empirical formula given in Eq. (34) provides a more adequate description of the velocity factor, Q_J , in the whole temperature range including the crossover region.

The quantum correction factor Γ evaluated exactly for the symmetric Eckart barrier is compared in Table I to those values Γ_1^* and Γ_2^* obtained with the approximate velocity factors Q_1 and Q_2 , respectively. The value Γ^* is defined by setting the velocity factor equal to one for all temperatures.⁴ Even at the lowest studied temperatures the last approximation reproduces correctly the order of magnitude of Γ . The six values corresponding to Γ_1^* at temperatures between 503 K and 125 K agree closely with those already reported in the literature,^{4,9} the mean deviation between both sets of data being about 1%. In Tables II and III, the exact values of J and K are compared to two sets of data derived from the path-integral centroid simulations for some selected temperatures. One set, J_2^* and K_2^* , is obtained with the approximation Q_2 to the velocity factor, and the other set, J^* and K^* ,

TABLE III. Mean transmission coefficient of the symmetric Eckart barrier. K is the exact value, $K^* = \text{erfc}(\beta F^*)^{1/2}$, and $K_2^* = Q_2^2 K^*$.

T (K)	K	K_2^*	K^*
10000	0.45	0.45	0.45
503.27	1.17×10^{-3}	1.11×10^{-3}	1.04×10^{-3}
301.96	4.45×10^{-5}	4.06×10^{-5}	3.53×10^{-5}
188.73	9.82×10^{-7}	8.23×10^{-7}	5.92×10^{-7}
125.82	4.95×10^{-8}	3.96×10^{-8}	1.86×10^{-8}
81.97	5.47×10^{-9}	3.77×10^{-9}	1.03×10^{-9}
48.78	9.68×10^{-10}	8.91×10^{-10}	1.41×10^{-10}
26.60	2.57×10^{-10}	1.73×10^{-10}	0.16×10^{-10}

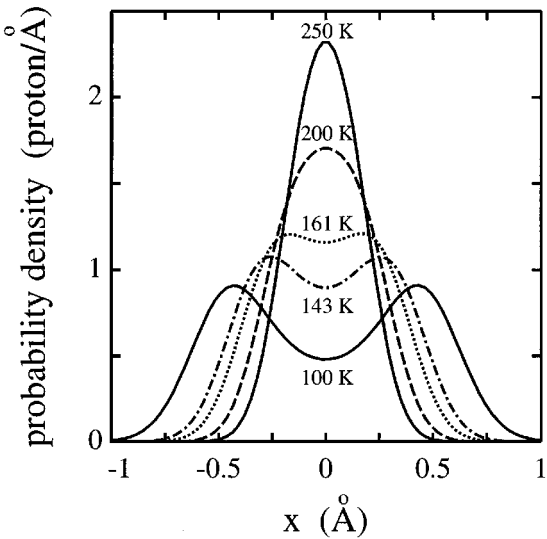


FIG. 7. Probability-density distribution of the proton as a function of temperature for PI centroid simulations with the centroid fixed at the barrier top. Note that below the crossover temperature at about 140 K, the distribution function for the quantum particle is bimodal. Above the crossover temperature the distribution function is unimodal.

is calculated by setting the velocity factor equal to one at all temperatures. It is remarkable that J^* provides an estimate to J that is never in error by more than a factor of 5, considering that J varies over 10 orders of magnitude in the studied temperature range. The error in K^* is about two times larger at the lowest temperatures. The values of J_2^* and K_2^* show that the approximate velocity factor Q_2 provides an accurate correction in the whole temperature range. The velocity factor Q_1 gives an accurate approximation as well, though slightly less accurate than Q_2 .

In previous investigations,^{2,12} a qualitative criterion has been applied to identify the quantum–classical crossover. It is related to the temperature dependence of the probability density, $\rho(x)$, obtained for a quantum particle in an equilibrium PI simulation with the centroid of the quantum paths fixed at the barrier top. It is expected that above the crossover temperature the equilibrium distribution of thermal paths leads to a unimodal $\rho(x)$ distribution. The predominance of quantum tunneling below the crossover temperature should be reflected by a qualitative change in the equilibrium distribution of thermal paths, with an increasing statistical weight for those paths that spread at both sides of the barrier, leading to a bimodal distribution of the probability density $\rho(x)$. The function $\rho(x)$ for several selected temperatures is shown in Fig. 7. The gradual change from a unimodal to a bimodal distribution is clear, and the appearance of a bimodal distribution correlates precisely with the crossover temperature found at 140 K.

V. SUMMARY

In the present work, I have applied the path-integral centroid approach to study several properties of a steady state associated to the transmission of a flux of protons in a sym-

metric Eckart barrier. In the present study we cover temperatures down to 10 K. The classical–quantum crossover for the barrier crossing was found at 140 K.

A result of the present numerical investigation is that the internal energy associated to the fixed centroid simulation has a plain physical meaning as it closely reproduces the difference between the mean kinetic energy of the transmitted and incident protons as a function of temperature. Moreover, the thermodynamical relation between this internal energy and the effective classical potential provides us with a new algorithm for computing effective-potential barriers, which is the essential quantity of the path-integral centroid approach to these kinds of problems. A simple expression for the velocity (pre-exponential) factor for this barrier-crossing problem is presented, that works better than previous approaches in the whole studied temperature range, above and below the crossover temperature.

The generalization of the results found in this work to other potential barriers is an open question. The main points to be analyzed are (i) an extension of the one-particle problem to many-body Hamiltonians; (ii) the study of asymmetric barriers; and (iii) the study of barrier crossing problems with potentials having bound states. These points will be the subject of future investigations.

ACKNOWLEDGMENTS

I thank J. C. Noya for helpful discussions and C. P. Herrero for critically reading the manuscript. This work was supported by the DGICYT (Spain) under Contract No. PB93-1254.

¹R. P. Feynman and A. R. Hibbs, *Quantum Mechanics and Path Integrals* (McGraw-Hill, New York, 1965).

²M. J. Gillan, *J. Phys. C Solid State Phys.* **20**, 3621 (1987).

³M. J. Gillan in *Computer Modelling of Fluids, Polymers, and Solids*, edited by C. R. A. Catlow, S. C. Parker, and M. P. Allen (Kluwer, Dordrecht, 1990).

⁴G. A. Voth, D. Chandler, and W. H. Miller, *J. Chem. Phys.* **91**, 7749 (1989).

⁵G. A. Voth, *Adv. Chem. Phys.* **93**, 135 (1996); *J. Phys. Chem.* **97**, 8365 (1993).

⁶G. A. Voth, *Chem. Phys. Lett.* **170**, 289 (1990).

⁷J. Cao and G. A. Voth, *J. Chem. Phys.* **99**, 10070 (1993); **100**, 5093, 5106 (1994); **101**, 6157, 6168, 6184 (1994).

⁸D. E. Makarov and M. Topaler, *Phys. Rev. E* **52**, 178 (1995).

⁹J. Cao and G. A. Voth, *J. Chem. Phys.* **105**, 6856 (1996).

¹⁰M. J. Gillan, *Philos. Mag. A* **58**, 257 (1988); *Phys. Rev. Lett.* **58**, 563 (1987).

¹¹Y. Sun and G. A. Voth, *J. Chem. Phys.* **98**, 7451 (1993).

¹²T. R. Mattsson and G. Wahnström, *Phys. Rev. B* **51**, 1885 (1995).

¹³C. P. Herrero, *Phys. Rev. B* **55**, 9235 (1997).

¹⁴J. C. Noya, C. P. Herrero, and R. Ramírez, *Phys. Rev. Lett.* (in press).

¹⁵N. Alberding, R. H. Austin, K. W. Beeson, S. S. Chan, L. Eisenstein, H. Frauenfelder, and T. M. Nordlund, *Science* **192**, 1002 (1976).

¹⁶J. Lobaugh and G. A. Voth, *Chem. Phys. Lett.* **198**, 311 (1992); *J. Chem. Phys.* **100**, 3039 (1994); **104**, 2056 (1996).

¹⁷J. B. Straus, A. Calhoun, and G. A. Voth, *J. Chem. Phys.* **102**, 529 (1995).

¹⁸R. P. McRae, G. K. Schenter, B. C. Garrett, G. R. Haynes, G. A. Voth, and G. C. Schatz, *J. Chem. Phys.* **97**, 7392 (1992).

¹⁹C. Eckart, *Phys. Rev.* **35**, 1303 (1930).

²⁰R. P. Bell, *The Tunnel Effect in Chemistry* (Chapman and Hall, London, 1980).

²¹H. S. Johnston, *Gas Phase Reaction Rate Theory* (Ronald, New York, 1966); H. S. Johnston and D. Rapp, *J. Am. Chem. Soc.* **16**, 1 (1961).

²²S. Wolfram, *MATHEMATICA* (Addison-Wesley, Redwood City, 1991).

²³R. P. Feynman and H. Kleinert, *Phys. Rev. A* **34**, 5080 (1986); G. A. Voth, *ibid.* **44**, 5302 (1991); A. Cuccoli, A. Macchi, M. Neumann, V. Tognetti, and R. Vaia, *Phys. Rev. B* **45**, 2088 (1992); D. Acocella, G. K. Horton, and E. R. Cowley, *ibid.* **51**, 11406 (1995).

²⁴H. Kleinert, *Pfadintegrale* (Wissenschaftsverlag, Mannheim, 1993).

²⁵R. Ramírez (unpublished).